

Notes: Li & Li (RES, 2025)

Firm Quality Dynamics and the Slippery Slope of Credit Intervention

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1 Big picture

- **Question.** What are the long-run consequences of direct government and central-bank credit support to nonfinancial firms in crises, and why can intervention grow from one crisis to the next?
- **Core mechanism.** Firms differ in productivity. High-productivity firms have more valuable capital, so they want to spend more on survival in crises and, because they are less likely to strategically default, they can also borrow more from private creditors. Crises therefore have a *cleansing effect*: low-productivity firms exit disproportionately.
- **Policy distortion.** Government credit support relaxes firms' liquidity constraints, but it is not tailored to productivity. The policy is size-based rather than productivity-based, so it benefits low-productivity firms relatively more and dampens the cleansing effect.
- **Main trade-off.** More intervention preserves *capital quantity* but lowers *capital quality*. The government saves more firms and more production capacity today, but it leaves the economy with a worse firm-quality distribution tomorrow.
- **Dynamic implication.** Because the post-crisis firm-quality distribution is biased downward, the economy enters the next crisis with weaker average productivity and lower private-sector financing capacity. That creates a *slippery slope*: future crises require larger interventions to contain the same output loss.
- **Main quantitative result.** Raising intervention from zero to a scale close to 30% of GDP saves about 8% of production capacity in the crisis, but reduces the quality improvement from roughly 9% to 4%.
- **Optimal policy result.** Welfare is hump-shaped in the intervention scale. The welfare-maximizing intervention is smaller than what a policymaker would choose if they only looked at output preservation and ignored endogenous firm-quality dynamics. In the paper's quantitative exercise, that "pseudo-optimal" policy is almost twice as large as the true optimum.
- **Persistence result.** In the forward-propagation exercise, moving intervention from zero to the paper's benchmark lowers the share of high-type firms by about 4 percentage points even ten years later and raises the intervention needed in the next crisis by roughly 7 cents per current dollar of support per unit of capital.
- **Why the paper matters.** Much of the crisis-policy literature treats the cost of direct lending as an exogenous deadweight loss. This paper opens that black box and shows an endogenous dynamic cost: intervention changes the composition of firms and thereby changes the future need for intervention.

2 Data and measurement (institutional context and key objects)

2.1 Economic environment and state variables

- **Agents.** A representative household, a government, and firms.
- **Firm types.** There are two firm types, H and L , with productivities $A^H > A^L$.
- **State variables.** The aggregate state is summarized by total capital K_t and the fraction of high-type capital ω_t . Quantity and quality are therefore distinct state variables.

- **Normal times.** Firms invest and grow capital according to Tobin’s q . There are no financing constraints outside crises.
- **Crises.** Crises arrive as a Poisson process. In a crisis, a firm must finance survival spending to preserve productive capital.

2.2 Crisis severity and survival technology

- **Liquidity need.** Each firm draws a crisis-exposure parameter ζ from a common distribution $H(\zeta)$. Lower ζ means a more severe liquidity crisis.
- **Survival probability.** If a firm raises x units of financing per unit of capital, survival probability is $F(x + \zeta)$, with F increasing and concave.
- **Interpretation.** A more severe crisis both lowers the firm’s baseline survival probability and raises the marginal value of extra liquidity.
- **Important distinction.** The model separates *productivity* from *credit risk*. A firm’s type affects capital value and default incentives, while ζ affects crisis intensity. The two are related, but not identical.

2.3 Financial frictions and policy design

- **Private funding limit.** In crises, private-sector creditors can supply at most $\bar{d}$ units of financing per unit of capital.
- **Strategic default friction.** Surviving firms may strategically default and renegotiate. If they do, owners obtain β and private creditors recover $q^j - \beta$. The government is assumed to recover zero in strategic default states.
- **Why type matters.** Since low-type firms have less valuable capital, they are more willing to strategically default. That makes their endogenous debt capacity lower.
- **Government intervention.** The government lends directly to firms and caps borrowing at $\bar{g}$ per unit of capital. It charges market-based rates but does *not* tailor credit quantity to productivity.
- **Institutional mapping.** The paper interprets this as a stylized representation of facilities like PMCCF, SMCCF, MSLP, and PPP, where eligibility or quantity limits were tied to operating scale, liabilities, or payroll rather than firm productivity.

2.4 Calibration and measurement discipline

- **Model size.** The quantitative version has 12 parameters.
- **Directly set from data.** Crisis frequency $\lambda = 0.07$, depreciation/exit $\delta = 0.2$, real discount rate $r = 0.066$, entry rate $\eta = 0.062$, and benchmark intervention $\bar{g} = 0.14$.
- **Moment-matched parameters.** The model matches the average investment-capital ratio, average GDP drop in crises, the effect of intervention on firm survival, average creditor recovery, output-capital ratio, TFP dispersion, and private debt-GDP ratio.

- **Headline calibration values.** Table 1 reports $\theta = 9.2$, $\lambda_F = 5.8$, $\ell_\zeta = 0.16$, $\beta = 0.43$, $A^H = 0.57$, $A^L = 0.15$, and $\bar{d} = 0.25$.
- **Empirical discipline.** The benchmark intervention level is chosen to match roughly 30% of GDP in U.S. COVID-era firm support, which maps into $\bar{g} \approx 0.14$ because the calibrated output-capital ratio is about 0.45.

3 Models and empirical specifications (all equations + interpretation)

3.1 Preferences, production, and the meaning of firm quality

Households are risk-neutral with utility

$$\mathbb{E} \left[\int_0^\infty e^{-rt} dc_t \right]. \quad (1)$$

Aggregate output is

$$Y_t = A^H K_t^H + A^L K_t^L = \left(A^H \omega_t + A^L (1 - \omega_t) \right) K_t, \quad (2)$$

with quality share and total capital given by

$$\omega_t = \frac{K_t^H}{K_t^H + K_t^L}, \quad K_t = K_t^H + K_t^L. \quad (3)$$

- **Interpretation.** Output depends on both *quantity* (K_t) and *quality* (ω_t). This separation is the backbone of the paper.
- **Why quality matters.** A policy can preserve aggregate capital but still worsen the composition of firms, lowering future productivity.

3.2 Investment in normal times: Tobin's q

A type- j firm chooses investment ι_t^j by solving

$$\max_{\iota_t^j} q_t^j \iota_t^j k_t - \Phi(\iota_t^j, k_t), \quad \Phi(\iota_t^j, k_t) = \left(\iota_t^j + \frac{\theta}{2} (\iota_t^j)^2 \right) k_t. \quad (4)$$

The optimal investment rule is

$$\iota_t^j = \frac{q_t^j - 1}{\theta}. \quad (5)$$

- **Interpretation.** Higher capital value raises investment. If $q^H > q^L$, high-type firms grow faster even outside crises.
- **Economic role.** Tobin's q links crisis policy to normal-time capital accumulation because interventions affect future survival and hence the value of capital.

3.3 Type-specific capital dynamics

Crises arrive with Poisson intensity λ . Type-specific capital evolves as

$$dK_t^H = K_t^H(\iota_t^H - \delta)dt - K_t^H \left[\int \left(1 - F(x_t^H(\zeta) + \zeta)\right) dH(\zeta) \right] dN_t, \quad (6)$$

$$dK_t^L = K_t^L(\iota_t^L - \delta)dt - K_t^L \left[\int \left(1 - F(x_t^L(\zeta) + \zeta)\right) dH(\zeta) \right] dN_t + \eta K_t dt. \quad (7)$$

- **Interpretation.** In normal times, capital grows by investment net of depreciation. In crises, some capital is destroyed because firms fail to survive.
- **Why the ηK_t term appears.** The authors add entry of low-type firms to keep the long-run quality distribution stationary.

3.4 The first-best crisis-survival problem

A planner would choose survival spending x to solve

$$\max_x F(x + \zeta)q_t^j - x. \quad (8)$$

The first-order condition is

$$F'(x_t^{*j}(\zeta) + \zeta)q_t^j = 1. \quad (9)$$

Define $\bar{\zeta}_t^j$ by

$$F'(\bar{\zeta}_t^j)q_t^j = 1, \quad (10)$$

so the first-best financing rule is

$$x_t^{*j}(\zeta) = (\bar{\zeta}_t^j - \zeta)_+. \quad (11)$$

- **Interpretation.** A firm spends until the marginal gain in survival probability times capital value equals the marginal resource cost.
- **Key comparative static.** Since $q_t^H > q_t^L$, high-type firms have $\bar{\zeta}_t^H > \bar{\zeta}_t^L$ and therefore want to spend more on survival.
- **First cleansing channel.** Even without default frictions, high-type firms value survival more and therefore have a higher target level of liquidity.

3.5 Competitive lending and the basic funding constraint

Given a crisis draw ζ , the firm chooses debt to solve

$$x_t^j(\zeta) = \arg \max_{0 \leq x \leq \bar{d} + \bar{g}} F(x + \zeta) \left[q_t^j - (1 + r_t^j(\zeta, x))x \right]. \quad (12)$$

Private creditors set rates from the break-even condition

$$F(x + \zeta)(1 + r_t^j(\zeta, x))x = x. \quad (13)$$

Without strategic default, the equilibrium spending rule becomes

$$x_t^j(\zeta) = \min\{x_t^{*j}(\zeta), \bar{d} + \bar{g}\}. \quad (14)$$

- **Interpretation.** With only a supply-side credit cap, government intervention is unambiguously helpful because it moves spending toward the first best.
- **Economic lesson.** If the only friction were insufficient private liquidity, the planner would simply choose a large enough $\bar{g}$.

3.6 Strategic default and endogenous debt capacity

Strategic default occurs when the owner prefers renegotiation value β to repayment. The threshold ζ_t^j solves

$$q_t^j - [1 + r_t^j(\zeta_t^j, x_t^j(\zeta_t^j))]x_t^j(\zeta_t^j) = \beta. \quad (15)$$

For firms that strategically default, private creditors cap lending at

$$\hat{d}_t^j(\zeta) = F(\hat{d}_t^j(\zeta) + \zeta + \bar{g})(q_t^j - \beta). \quad (16)$$

The equilibrium financing rule is then

$$x_t^j(\zeta) = \mathbf{1}_{\{\zeta \geq \zeta_t^j\}} \min\{(\bar{\zeta}_t^j - \zeta)_+, \bar{d} + \bar{g}\} + \mathbf{1}_{\{\zeta < \zeta_t^j\}} \left(\min\{\hat{d}_t^j(\zeta), \bar{d}\} + \bar{g} \right). \quad (17)$$

- **Interpretation.** Firms that repay debt stop at the first-best or the funding ceiling. Firms that strategically default borrow as much as possible.
- **Second cleansing channel.** High-type firms have lower default thresholds ($\zeta_t^H < \zeta_t^L$) and higher endogenous debt capacity ($\hat{d}_t^H(\zeta) > \hat{d}_t^L(\zeta)$), so they can borrow more in crises.
- **Why this matters.** This demand-side friction is exactly where government intervention becomes distortionary. A uniform credit facility weakens the private sector's productivity-based discipline.

3.7 Under-spending, over-spending, and crowding in

The paper's key efficiency result is sign-based rather than a separate structural equation:

$$\frac{\partial \hat{d}_t^j(\zeta)}{\partial \bar{g}} > 0, \quad \frac{\partial r_t^j(\zeta)}{\partial \bar{g}} \leq 0. \quad (18)$$

- **Interpretation.** Government lending does not just add liquidity directly; it also crowds in private funding by raising survival probabilities and lowering spreads.

- **Good effect.** If a firm is under-spending relative to the first best, extra intervention raises welfare.
- **Bad effect.** If a strategically defaulting firm is already borrowing beyond the first-best survival level, more intervention worsens over-spending.
- **Core proposition.** Small intervention can help, but sufficiently large intervention becomes welfare-reducing because over-spending dominates.

3.8 Capital value and stationary equilibrium

The owner's value from a crisis draw ζ is

$$\pi_t^j(\zeta) = \left(F(x_t^j(\zeta) + \zeta)q_t^j - x_t^j(\zeta) \right) \mathbf{1}_{\{\zeta \geq \zeta_t^j\}} + F(x_t^j(\zeta) + \zeta)\beta \mathbf{1}_{\{\zeta < \zeta_t^j\}}. \quad (19)$$

Capital value solves

$$r = \mathbb{E}_t \left[\frac{dq_t^j/dt}{q_{t-}^j} \right] + \frac{A^j - \phi(\iota_{t-}^j)}{q_{t-}^j} + (\iota_{t-}^j - \delta) + \lambda \frac{\int_0^\infty \pi_t^j(\zeta) dH(\zeta) - q_{t-}^j}{q_{t-}^j}. \quad (20)$$

- **Interpretation.** Tobin's q capitalizes normal-time cash flows, normal-time investment growth, and crisis payoffs.
- **Main equilibrium result.** The stationary equilibrium is unique and has $q^H > q^L$, which makes high-type firms both more valuable and faster growing.
- **Why this is central.** All the crisis thresholds – desired spending, default incentives, and debt capacity – are functions of q^j .

3.9 Aggregate dynamics: quantity and quality

Total capital evolves as

$$\frac{dK_t}{K_{t-}} = \mu_t^K(\omega_{t-})dt + \Delta_t^K(\omega_{t-})dN_t, \quad (21)$$

where

$$\mu_t^K(\omega_{t-}) = -\delta + \omega_{t-}\iota_{t-}^H + (1 - \omega_{t-})\iota_{t-}^L + \eta, \quad (22)$$

$$\Delta_t^K(\omega_{t-}) = \omega_{t-}\kappa_t^H + (1 - \omega_{t-})\kappa_t^L - 1, \quad \kappa_t^j = \int F(x_t^j(\zeta) + \zeta) dH(\zeta). \quad (23)$$

Firm quality evolves as

$$\begin{aligned} d\omega_t = & \omega_{t-}(1 - \omega_{t-}) \left(\iota_{t-}^H - \iota_{t-}^L - \frac{\eta}{1 - \omega_{t-}} \right) dt \\ & + \left(\frac{\omega_{t-}\kappa_t^H}{\omega_{t-}\kappa_t^H + (1 - \omega_{t-})\kappa_t^L} - \omega_{t-} \right) dN_t. \end{aligned} \quad (24)$$

- **Interpretation.** The drift of K_t is normal-time growth; the jump term is crisis destruction. The drift of ω_t reflects faster growth by high-type firms offset by low-type entry; the jump term captures the crisis cleansing effect.
- **Cleansing condition.** Crises increase firm quality when $\kappa_t^H > \kappa_t^L$.
- **Main comparative static.** The paper shows

$$\frac{\partial \Delta_t^K}{\partial \bar{g}} > 0, \quad \frac{\partial \Delta_t^\omega}{\partial \bar{g}} < 0,$$

so intervention preserves more capital but weakens the cleansing effect.

3.10 Welfare and the optimal intervention rule

Let welfare per unit of capital be $W(\omega_t)$, so total welfare is $W(\omega_t)K_t$. The stationary Bellman equation is

$$\begin{aligned} rW(\omega) = & \omega A^H + (1 - \omega)A^L - (\omega \iota^H + (1 - \omega)\iota^L) + W(\omega)\mu^K(\omega) + W'(\omega)\mu^\omega(\omega) \\ & - \lambda I(\omega) + \lambda \left[W(\omega + \Delta^\omega(\omega))(1 + \Delta^K(\omega)) - W(\omega) \right]. \end{aligned} \quad (25)$$

The planner's intervention problem is

$$g^*(\omega_0) = \arg \max_{\bar{g}} W(\omega_0; \bar{g}). \quad (26)$$

The paper also defines a “pseudo-optimal” rule that ignores quality dynamics,

$$\tilde{g}^*(\omega_0) = \arg \max_{\bar{g}} \widetilde{W}(\omega_0; \bar{g}), \quad (27)$$

where $\widetilde{W}$ incorrectly treats $\Delta^\omega(\omega; \bar{g})$ as if it were unaffected by policy.

- **Interpretation.** Welfare trades off current output preservation against future firm-quality distortion.
- **Policy lesson.** Ignoring the effect of intervention on ω leads to too much intervention, because the policymaker internalizes quantity but not future quality.

4 Identification and instruments (what makes the quantitative exercise credible)

4.1 This is not reduced-form identification

- The paper is a structural, theory-driven quantitative exercise rather than an econometric identification paper.
- The relevant notion of identification is **mechanism identification**: the model isolates a precise wedge between private and public credit allocation.

- Private creditors price and ration credit in a productivity-sensitive way because default incentives differ by type. The government follows market rates but not productivity when setting credit quantity.
- That single asymmetry is what generates both the quantity-quality trade-off and the slippery slope.

4.2 What disciplines the wedge between high- and low-type firms

- **Capital value discipline.** Because $q^H > q^L$, high-type firms want to preserve capital more and lose more from default.
- **Debt-capacity discipline.** Low-type firms have stronger strategic-default incentives, so their endogenous private debt capacity is lower.
- **Policy design discipline.** The government's cap $\bar{g}$ is uniform per unit of capital, which is crucial. If policy could perfectly condition on productivity, the paper's distortion would disappear.
- **Credit-risk distinction.** The model does not equate productivity with credit risk one-for-one, which is important because many real-world programs screen on credit quality, not productivity.

4.3 Why Condition 1 is important quantitatively

- The key quantitative restriction is that q^H is high enough and q^L is low enough that: (i) type-H firms do not strategically default, (ii) type-L firms' first-best survival spending is below the total funding cap, and (iii) type-H firms still want to spend more than type-L firms around the type-L default region.
- Under this condition, type-H firms mainly suffer from *under-spending*, while type-L firms are the ones for which *strategic default and over-spending* become salient.
- This is exactly the region in which extra intervention disproportionately helps low-productivity firms and therefore weakens crisis cleansing.

4.4 Calibration discipline

- **Normal-time discipline.** Investment, output-capital ratio, entry, and discounting pin down the non-crisis side of the model.
- **Crisis discipline.** Crisis frequency, GDP drop, creditor recovery, and the survival effect of intervention pin down the crisis side.
- **Cross-sectional discipline.** TFP dispersion and private debt-GDP ratio pin down the gap between H and L firms and the tightness of market funding.
- **Policy scale discipline.** The benchmark $\bar{g}$ is tied to observed COVID-era interventions rather than chosen purely for convenience.

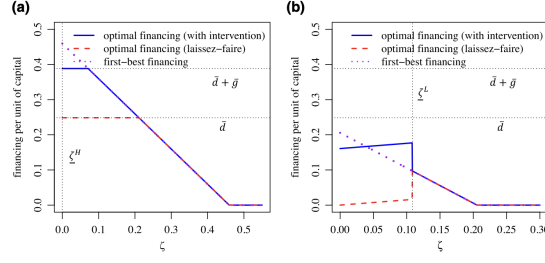


FIGURE 1
Optimal firm financing. (a) Type- H firm financing. (b) Type- L firm financing
Note: This figure illustrates the optimal firm financing in different scenarios, under the calibrated model parameters that we discuss in Section 3. The solid line shows the optimal financing under credit intervention. The dashed line shows the optimal financing without credit intervention ($\bar{g} = 0$). The dotted line shows the first-best financing amount without funding supply- and demand-side frictions

4.5 Why the slippery-slope result is not only an expectations result

- To isolate the mechanism, the paper shuts down agents' expectations of future intervention when studying inter-crisis pass-through.
- Concretely, the authors solve capital values q^H and q^L under $\bar{g} = 0$, so the dynamic result is driven by *forward propagation* only: today's intervention changes tomorrow's firm-quality distribution.
- The slippery slope survives even under this conservative exercise, which strengthens the claim that the mechanism is not merely a rational-expectations artifact.
- The appendix then shows that the same pattern also appears in a fully dynamic policy environment with rational expectations.

4.6 Main limitations to keep in mind

- The model has only two firm types, so it is best read as a clean approximation rather than a literal description of the firm distribution.
- Households are risk-neutral, and the model abstracts from aggregate-demand effects, banking-sector balance-sheet problems, sovereign risk, and political economy.
- Government lending uses market-based rates, but government recovery in default is zero. This sharp assumption helps generate the distortion and fits the authors' interpretation of abuse in emergency programs, but it is still stylized.
- The core normative conclusion is therefore most compelling as a *new mechanism* to add to crisis-policy analysis, not as the final word on the exact optimal size of lending facilities.

5 Tables and figures

5.1 Figure 1: Optimal firm financing

Read:

- Panel (a) shows type- H financing. High-type firms never strategically default in the calibrated region, so government intervention simply relaxes under-spending up to the first-best and then flattens out.

- Panel (b) shows type-L financing. Low-type firms can strategically default, so financing rises with intervention even after the first-best benchmark is crossed. That is where over-spending appears.
- This is the cleanest visual for the paper’s asymmetry: intervention mostly reduces under-spending for type-H firms, but increasingly fuels over-borrowing by type-L firms.

5.2 Tables 1 and 2: Parameter values and moment matching

Read:

- Table 1 shows the benchmark calibration. The central intervention parameter is $\bar{g} = 0.14$, which the authors map to support on the order of 30% of GDP.
- Table 2 shows the model matches its targets exactly or very closely: investment-capital ratio 0.1, crisis GDP drop -9.3% , survival effect of intervention 10%, creditor recovery 49%, output-capital ratio 45%, TFP ratio 3.7, and private debt-GDP ratio 36%.
- These tables matter because the later welfare and pass-through results are not purely illustrative; they are disciplined by crisis and firm-level moments.

TABLE 1
Parameter values

Parameter	Description	Value	Data Source or Targeted Moment
θ	Investment cost	9.2	Average investment/capital ratio
λ_F	Survival rate	5.8	Average GDP drop in crises
l_C	Average ζ	0.16	Impact of credit intervention on firm survival
λ	Crisis frequency	0.07	Crisis frequency in the data
β	Debt restructuring rent	0.43	Average creditor recovery rate
δ	Capital depreciation rate	0.2	Capital depreciation and firm exit rate
r	Real discount rate	0.066	Average real bond return plus equity premium
A^H	Productivity of type-H firms	0.57	Average output-to-capital ratio
A^L	Productivity of type-L firms	0.15	TFP inter-quartile ratio
η	Entry rate of new firms	0.062	Firm entry rate
$\bar{d}$	Private-sector credit availability	0.25	Private-sector debt/GDP ratio
$\bar{g}$	Government credit support	0.14	COVID-19 credit support in the U.S.

Note: This table shows the parameter values in our quantitative analysis and the corresponding calibration targets. Five model parameters are directly set by the observed data, including λ , δ , r , η , and $\bar{g}$. The rest of parameters, θ , λ_F , l_C , β , A^H , A^L , and $\bar{d}$ are solved to match the moment targets.

TABLE 2
Moment matching

Moment Description	Model	Data
Average investment-to-capital ratio	0.1	0.1
Average GDP drop in crises	-9.3%	-9.3%
Average impact of credit intervention on firm survival likelihood	10%	10%
Average creditor recovery rate	49%	49%
Average output-to-capital ratio	45%	45%
TFP ratio between 90% and 10% percentiles	3.7	3.7
Average private-sector debt/GDP ratio	36%	36%

Note: We report the moment matching results for calibrating θ , λ_F , l_C , β , A^H , A^L , and $\bar{d}$.

5.3 Figure 2: Credit intervention impact — capital quantity vs. quality

Read:

- Panel (a) shows that larger intervention makes the crisis less destructive for total capital. Moving from zero support to the benchmark meaningfully shrinks the percentage fall in K_t .
- Panel (b) shows the flip side: the same increase in intervention sharply reduces the jump in ω_t , that is, it weakens the cleansing effect.
- This figure is the paper in one picture: quantity goes up, quality goes down.

5.4 Figure 3: Credit intervention and firm financing

Read:

- Panel (a) compares the borrowing of a type-H firm and a strategic-defaulting type-L firm as $\bar{g}$ rises. High-type borrowing flattens once the first-best is reached; low-type borrowing keeps rising.
- Panel (b) decomposes type-L borrowing into private and government sources. Government support directly adds funding and also crowds in private funding by improving survival probabilities.

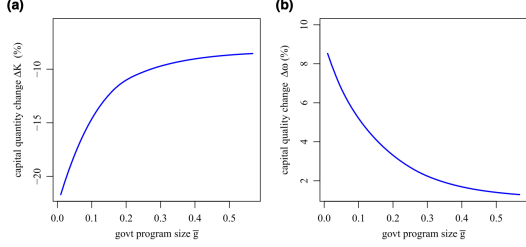


FIGURE 2

Credit intervention impact: Capital quantity vs. quality. (a) Credit intervention and capital quantity. (b) Credit intervention and capital quality

Note: This figure illustrates how $\bar{g}$ affects ΔK and Δa in a crisis. The calculation requires the pre-crisis α_0 – which we set to the average value $\bar{\alpha}$ defined in Section 3.1

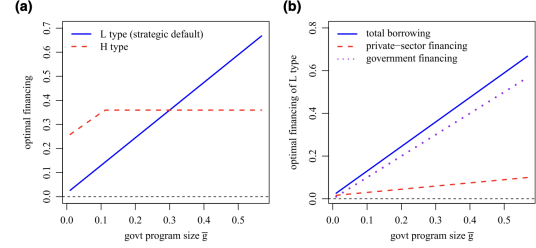


FIGURE 3

Credit intervention and firm financing. (a) Firm financing: $x_L^H(L)$ and $x_L^H(L)$. (b) Decomposing type-L firm financing

Note: This figure illustrates how intervention affects the optimal amount of financing by H and L type firms, accounting for the endogenous responses of q^H and d^L . In both panels, we choose $\zeta = 0.1$, which is a case of strategic default for L-type firms. In panel (b), we decompose the actual borrowing of L-type firms into private-sector financing (which reaches the endogenous limit d^L) and government financing (which reaches the limit $\bar{g}$). As $\bar{g}$ changes, all equilibrium variables change accordingly, including capital values

- This figure explains why the policy distortion is strong: intervention helps low-type firms on both the extensive and intensive margins of financing.

5.5 Figure 4: Credit intervention and welfare

Read:

- Panel (a) shows average firm quality $\bar{\omega}(\bar{g})$ is decreasing in intervention.
- Panel (b) shows welfare is hump-shaped. Small interventions are welfare-improving, but excessive intervention becomes harmful.
- The peak occurs at a program size of roughly 22% of GDP, which is below the paper's benchmark intervention scale motivated by the COVID episode.

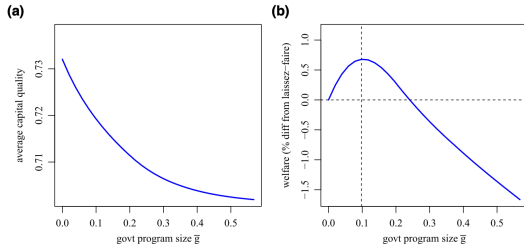


FIGURE 4

Credit intervention and welfare. (a) Average capital quality. (b) Welfare improvement

Note: In panel (a), we plot the average firm quality $\bar{\omega}$ as a function of government intervention $\bar{g}$. For each $\bar{g}$, we solve the model again and calculate the average of simulated α_0 as $\bar{\alpha}(\bar{g})$. In panel (b), we show the welfare difference $W(\bar{\alpha}(\bar{g}), \bar{g})/W(\bar{\alpha}(0), 0) - 1$ as a function of government intervention $\bar{g}$

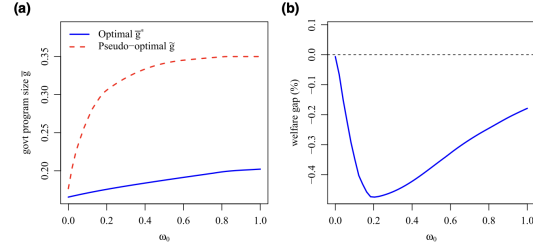


FIGURE 5

Optimal intervention and welfare. (a) Optimal government intervention. (b) Difference in optimal welfare

Note: In panel (a), we plot the optimal intervention $\bar{g}^*(\omega_0)$ as a function of the initial state ω_0 , and the pseudo-optimal policy $\tilde{g}(\omega_0)$ that ignores the impact of intervention on firm quality change, treating $\Delta^H(\omega; \bar{g})$ as $\Delta^H(\omega; 0)$ for all $\bar{g}$. In panel (b), we show the percentage decline of social welfare due to using the pseudo-optimal policy instead of the optimal policy

5.6 Figure 5: Optimal intervention and pseudo-optimal intervention

Read:

- Panel (a) plots the truly optimal $g^*(\omega_0)$ and the pseudo-optimal $\tilde{g}^*(\omega_0)$ that ignores quality dynamics.
- The pseudo-optimal policy is almost twice as large across much of the state space, especially when the economy is not dominated by one type or the other.
- Panel (b) shows the welfare gap from using the pseudo-optimal rule. The welfare loss is largest at intermediate initial quality shares, where the policy trade-off is sharpest.

5.7 Figure 6: Intervention pass-through across crises

Read:

- Panel (a) shows that a larger intervention today leaves the economy with a lower share of high-type firms ten years later.
- Panel (b) translates that into future policy: the intervention needed in the next crisis rises with the current intervention scale.
- This is the paper’s main dynamic fact. The effect does not disappear quickly, and the implied inter-crisis pass-through is economically meaningful.

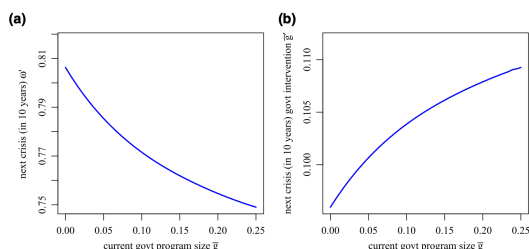


FIGURE 6
Intervention pass-through across crises. (a) Firm quality entering the next crisis. (b) Required intervention in the next crisis

Note: We show how $\bar{g}$ in the current crisis affects capital quality entering the next crisis, α' , and intervention needed, $\bar{g}'$, in the next crisis to contain output drop within -10% . The next crisis happens ten years after the current one. Agents expect no intervention (pass-through is only due to forward propagation). The current crisis happens at α equal to the average value of α_t in the laissez-faire economy

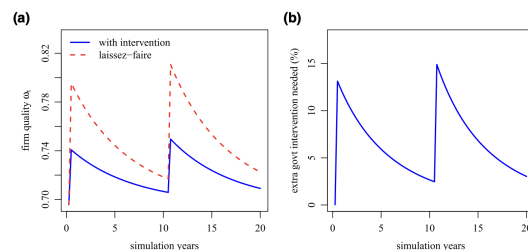


FIGURE 7
The slippery slope of intervention: Simulated paths. (a) Average capital quality. (b) Additional intervention needed

Note: We compare the quarterly simulation of two economies, the laissez-faire economy and the intervened economy without agents’ expectation of intervention, where one crisis happens in Q1 of the first year and another crisis in Q1 of the tenth year. Both simulations start at the average α_0 in the laissez-faire economy. In Panel (a), we plot the path of capital quality, α_t . In Panel (b), for both economies, we calculate the amount of intervention required to contain output drop within -10% if a crisis happens over the next instant, and then we plot the percentage increase from the laissez-faire economy to the intervened economy

5.8 Figure 7: Simulated slippery-slope paths

Read:

- Panel (a) simulates two crises, one immediately and one ten years later. The intervention economy always has a lower quality path than the laissez-faire benchmark, with the wedge widening after each crisis.
- Panel (b) shows the additional support the intervention economy would need relative to the laissez-faire economy if another crisis hit at each point in time.
- The key pattern is persistence: the extra intervention need spikes after crises but remains positive for years.

5.9 Figure 8: Alternative policy design — pure subsidy

Read:

- The authors compare their baseline loan policy with a policy in which the government subsidizes survival spending instead of demanding repayment.
- Even pure subsidies still distort firm quality, but welfare improves over a wider range of intervention sizes and the optimum shifts upward.
- Intuition: subsidy reduces private-credit strategic default incentives and crowds in “informed” private liquidity, so it dominates the baseline loan design in this model.

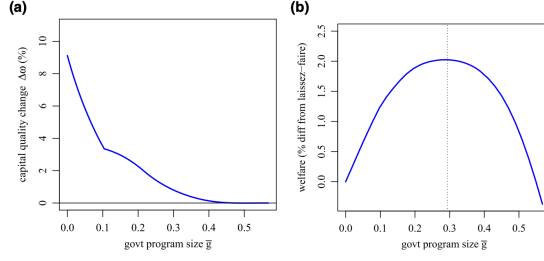


FIGURE 8
Impact of pure government subsidy: capital quality and welfare. (a) Credit intervention and capital quality. (b) Welfare improvement

Note: This figure illustrates how $\bar{g}$ (government intervention is pure subsidy) affects Δa and welfare. The calculation requires the pre-crisis ω_i , which we set to the average value $\bar{\omega}$. For each $\bar{g}$, we solve the model again and calculate the average of simulated ω_i as $\bar{\omega}(\bar{g})$. In panel (b), we show the welfare difference $W(\bar{\omega}(\bar{g}); \bar{g})/W(\bar{\omega}(0); 0) - 1$ as a function of government intervention $\bar{g}$.

6 Short summary

- **Conclusion.** Direct crisis lending to firms can be useful in the short run but distortive in the long run because it weakens the cleansing effect of crises.
- **Intuition.** Private creditors naturally favor high-productivity firms because those firms are less likely to strategically default. A uniform public credit facility compresses that discipline and keeps too many low-productivity firms alive.
- **Quantitative lesson.** The policy question is not “lend or do not lend.” It is how much to lend once one recognizes that intervention changes the future firm distribution.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Credit intervention faces a quantity-quality trade-off. It preserves capital quantity in crises but lowers capital quality by saving low-productivity firms disproportionately.
- **Mechanism.** The distortion comes from the interaction of two features: private creditors screen firms through strategic-default incentives, while the government supplies credit through a uniform size-based rule.
- **Dynamic implication.** Because intervention worsens the post-crisis quality distribution, the economy enters later crises with lower productivity and lower private financing capacity. This creates a slippery slope of future intervention.
- **Optimal policy implication.** The welfare-maximizing intervention is materially smaller than the policy one would choose by looking only at current output stabilization.
- **Quantitative punchline.** Ignoring endogenous firm-quality dynamics leads to severe over-intervention; in the paper’s benchmark, the implied policy is almost twice the optimum.

7.2 Economic intuition

- A crisis is not just a drop in output; it is also a reallocation event. Some firms survive and others do not.

- Private credit markets, despite being frictional, still allocate scarce crisis liquidity in a productivity-sensitive way because default incentives reveal which firms have more valuable continuation opportunities.
- Public intervention helps by overcoming funding shortages, but it hurts when it homogenizes access to liquidity across firms whose continuation values are very different.
- The long-run cost is therefore compositional: not all saved firms are equally valuable for the future path of the economy.

7.3 Takeaway

This paper's main contribution is to show that direct credit intervention has an *endogenous dynamic cost*. The distortion is not just a one-time subsidy or a fiscal burden; it operates through the evolution of firm quality itself. For reading the current policy debate, the lesson is that aggressive crisis lending can be justified, but only after recognizing that today's intervention changes the composition of tomorrow's firms and therefore changes the scale of tomorrow's rescue.